

HOVID CHLORAMPHENICOL CAPSULE

VICH03-1

Description

Green opaque and white opaque capsule with "HD" printed on one end and "CA 250" on the other end of the capsule.

Composition

Each capsule contains:

Chloramphenicol 250 mg per capsule.

Actions and Pharmacology

Chloramphenicol inhibits protein synthesis in bacteria and in cell-free systems by binding in a reversible manner to the 50 S ribosomal subunit. It suppresses the activity of peptidyl transferase and disorganizes the amino acid coding which catalyses peptide bond formation but does not prevent the ability of ribosomes to bind to and move along strands of messenger RNA. The overall effect involves uncoupling of two processes : ribosomal translocation and peptide bond synthesis.

Chloramphenicol possesses a fairly wide spectrum of anti-microbial activity. It is primarily bacteriostatic, although it may be bactericidal to certain species under some conditions. Among the bacteria inhibited by relatively low concentrations of the antibiotic in vitro are :

Enterobacter (Aerobacter) aerogenes; Escherichia coli; Klebsiella pneumoniae; Bordetella pertussis; Haemophilus influenzae; Pasteurella; Pseudomonas and Pseudomonas; Bacteroides; Salmonella typhi and other species; Proteus vulgaris; Neisseria; Shigella; Brucella; Vibrio cholerae.

Some streptococci and staphylococci are suppressed by higher concentrations. Actinomyces, Bacillus anthracis, Corynebacterium diphtheriae, Clostridium, Listeria, Bartonella and Leptospira are sensitive to moderate concentrations.

Resistant bacteria include Pseudomonas (except Pseudomonas pseudomallei), Mycobacteria and Nocardia. Fungi including yeasts and protozoa are resistant.

Pharmacokinetics**Absorption**

Following oral administration, chloramphenicol is rapidly and completely absorbed from the gastrointestinal tract. Simultaneous administration of food or antacids does not impair its absorption to any significant extent.

Blood concentration

Significant plasma concentrations are obtained within 30 minutes. Peak serum concentrations ranging from 8-20 µg/ml (mostly about 10µg/ml) are reached at 1 to 9 hours (majority in 2 hours) after a single oral dose of 1g.

Following administration of 500mg of chloramphenicol, serum levels were about 33% higher in female than in male patients. Oral doses of 500 mg every 6 hours usually maintains the blood concentration about 4µg/ml.

Half-life

The biological half-life of chloramphenicol is 1.05 to 3.5 hours and there is virtually no drug detectable in the blood after 12 to 18 hours. Serum half-life in adults after oral administration is 2 to 5 hours, which is increased in hepatic function impairment.

Distribution

Chloramphenicol is widely distributed in body fluids and tissues and readily reaches therapeutic concentrations in cerebrospinal fluid and brain. It enters the cerebrospinal fluid even in the absence of meningitis giving concentrations of about 30 to 50% of blood levels.

Clinical studies with young infants have indicated that transference of chloramphenicol from serum to cerebrospinal fluid was about twice as high in the acute inflammatory stage of meningitis ie 60% as in the normal state ie 30%. Furthermore, the passage of chloramphenicol to the cerebrospinal fluid has been shown to be independent of age.

Chloramphenicol administered in clinical doses has been shown to produce concentrations in cerebral tissue 9 times that of the blood level, and hence is sufficient to inhibit all usual intracerebral pathogens except Pseudomonas. However, the highest tissue concentration of chloramphenicol appear in the liver and kidneys. Chloramphenicol crosses the placenta and is secreted in the saliva, bile, milk, pleural fluid and ascites fluid.

Protein Binding

At peak blood levels, about 60% of the chloramphenicol is bound by plasma albumin.

Metabolism

Under conditions of normal hepatic function, about 92% of administered chloramphenicol is metabolized in the liver. Out of this 92%, about 90% undergoes conjugation to the glucuronide by glucuronyl transferase and 2% undergoes reduction to arylamines of undergoes deacetylation and dehalogenation.

In patients with extensive liver disease (eg. cirrhosis) and neonates, the capacity to conjugate chloramphenicol with glucuronic acid is reduced. In those patients, the dosage will have to be modified.

Excretion

Chloramphenicol is excreted in the urine and bile. Nearly 90% of the administered chloramphenicol is excreted in the urine during the first six hours after ingestion. Most of this is eliminated by tubular secretion as inactive glucuronide conjugates and arylamines.

5 to 10% of the chloramphenicol excreted in the urine is eliminated as the active compound and this is largely by glomerular filtration.

Approximately 3% of a dose of chloramphenicol is excreted in the bile in the active form but most of this is reabsorbed from the intestine so that only about 1% is excreted in the faeces and this is an inactive form.

BACK

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Indications

For treatment of:

- Typhoid fever and similar salmonella infections.
- *Haemophilus influenzae meningitis*.
- Severe anaerobic infections due to *Bacteroides fragilis*.
- Rickettsial infections which do not respond to treatment with other drugs.
- Lymphogranuloma-Psittacosis infections.
- Serious chest infections and situations where clinical assessment usually supplemented by laboratory studies indicates that no other antibiotic would suffice.

Contraindications

- Avoid in patients who have had previous toxic reactions to the drug.
- Use is not recommended in pregnancy and nursing mothers.
- It should not be used in trivial infections; or for prophylaxis.

Precautions

- Caution in patients with hepatic or renal function impairment.
- Regular blood counts should be performed as blood dyscrasias may occur.
- Stop administration and check with physician immediately if symptoms of grey baby syndrome occur in an infant or child.
- Repeated courses and prolonged treatment should be avoided.

Main Side/Adverse Effects

Bone marrow depression characterised by agranulocytosis, thrombocytopenia, aplastic anaemia; gastrointestinal symptoms including nausea, vomiting, diarrhoea, glossitis, stomatitis and enterocolitis; grey syndrome; optic and peripheral neuritis; hypersensitivity reactions.

Drug Interactions

- Effects and/or toxicity of oral anticoagulants, oral hypoglycaemics or phenytoin may be increased when used concurrently with chloramphenicol.
- Co-administration of bone-marrow depressants, for example, antineoplastics, colchicine, gold salts, oxyphenbutazone, penicillamine, phenylbutazone may increase the potential for toxic side effects.
- Avoid concurrent therapy with lincomycins or penicillins or paracetamol.

Pregnancy and Lactation

Not recommended in pregnancy and lactation.

Overdosage

Clinical features : Nausea, vomiting, diarrhoea, hepatitis, bone marrow depression.

Treat overdosage by emesis or gastric lavage, if appropriate; with the necessary symptomatic and supportive measures, if required.

Dosage and Administration

Adults : Oral, 500 mg every six hours or as directed.

Children : As directed by physician.

Note: In severe infections, doses up to 100 mg per kg of body weight daily may be used.

The information given here is limited. For further information consult your doctor or pharmacist.

Storage : Store below 25°C. Protect from moisture.

Presentation/Packing : Capsule 250 mg x 1000's, Blisters of 10 x 10's, 3 x 10's.

Product Registration holder: HOVID Bhd., 121, Jalan Tunku Abdul Rahman, 30010 Ipoh, Malaysia.

Manufactured by: HOVID Bhd., Lot 56442, 7 1/2 Miles, Jalan Ipoh/Chemor, 31200 Chemor, Malaysia.

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